

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277371

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Gebhardt; David R. et al.

Light Emitting Handrail

Abstract

A handrail that includes a cover, a first endcap, a second endcap, a support member, and a lighting element is provided. The support member includes a longitudinal axis, first and second ends along the longitudinal axis, first and second lateral sides, and first and second longitudinal sides. The first lateral side of the support member is configured to be secured against a mounting surface. The first endcap is configured to mate with the first end of the support member and the second endcap is configured to mate with the second end of the support member. The cover is placed over the second lateral side of the support member and engages the first longitudinal side and the second longitudinal side of the support member. The lighting element is configured to couple to the support member in a lighting channel and emit light toward the mounting surface.

Inventors: Gebhardt; David R. (Milwaukee, WI), Bennett; Matthew G. (New Berlin, WI)

Applicant: InPro Corporation (Muskego, WI)

Family ID: 96881002

Appl. No.: 18/593118

Filed: March 01, 2024

Publication Classification

Int. Cl.: E04F11/18 (20060101); F21S4/28 (20160101); F21V23/00 (20150101); F21V33/00 (20060101); F21Y113/00 (20160101); F21Y115/10 (20160101)

U.S. Cl.:

CPC E04F11/1808 (20130101); E04F11/1804 (20130101); F21S4/28 (20160101); F21V23/001 (20130101); F21V33/006 (20130101); E04F2011/1872 (20130101); F21Y2113/00 (20130101); F21Y2115/10 (20160801)

Background/Summary

BACKGROUND OF THE INVENTION

[0001] The present invention relates generally to the field of handrails. The present invention relates specifically to handrails having lighting systems capable of emitting light and/or providing ambient or accent room lighting.

SUMMARY OF THE INVENTION

[0002] One embodiment of the invention relates to a handrail configured to be installed on a mounting wall. The handrail includes an elongate support member having a first longitudinal end, a second longitudinal end, and a longitudinal axis extending through the first longitudinal end and the second longitudinal end. The support member includes a base member having a first side that faces toward the mounting wall when the support member is installed and a second side that faces away from the mounting wall when the support member is installed. The support member includes a channel formed in the first side and extending in a direction parallel to the longitudinal axis and a mounting arm extending from the first side of the base member, the mounting arm being configured to engage the mounting wall when the support member is installed. The handrail includes a cover configured to couple to the support member. The cover includes a substantially continuous surface configured to engage the first side of the base member at a first location that is beneath the mounting arm, cover the second side of the base member, and engage the base member at a second location that is above the mounting arm. The handrail includes a lighting element coupled to the base member within the channel such that light emitted by the lighting element is directed away from the second side of the base member. The base member is a separate, discontinuous component from the cover such that the cover can be coupled to and removed from the base member while the lighting element is coupled to the base member.

[0003] Another embodiment of the invention relates to a handrail configured to be installed on a mounting wall. The handrail includes an elongate support member having a first longitudinal end, a second longitudinal end, and a longitudinal axis extending through the first longitudinal end and the second longitudinal end. The support member includes a base member having a first side that faces toward the mounting wall when the support member is installed and a second side that faces away from the mounting wall when the support member is installed. The base member includes a channel that extends in a direction parallel to the longitudinal axis and a mounting arm extending from the first side of the base member. The mounting arm is configured to engage the mounting wall when the support member is installed. The handrail includes a cover configured to couple to the support member. The cover includes a substantially continuous surface configured to engage the first side of the base member at a first location that is beneath the mounting arm, cover the second side of the base member, and engage the base member at a second location that is above the mounting arm. The handrail includes a lighting element coupled to the base member within the channel. The lighting element includes a power cord for directing electrical energy to the lighting element from a power source. The handrail includes an endcap having a first endcap end and a second endcap end, the endcap coupled to the first longitudinal end of the support member at the first endcap end, and the endcap forming an arcuate shape that defines an arc between the first endcap end and the second endcap end. The power cord of the lighting element passes through the endcap from the lighting element to the mounting wall. An outer surface of the cover abuts and aligns substantially flush with an outer surface of the endcap.

[0004] Another embodiment of the invention relates to a handrail configured to be installed on a mounting wall. The handrail includes an elongate support member having a first longitudinal end, a second longitudinal end, and a longitudinal axis extending through the first longitudinal end and the second longitudinal end. The support member includes a base member having a first side that

faces toward the mounting wall when the support member is installed and a second side that faces away from the mounting wall when the support member is installed. The support member includes a channel formed in the first side and extending in a direction parallel to the longitudinal axis and a mounting arm extending from the first side of the base member, the mounting arm being configured to engage the mounting wall when the support member is installed. The handrail includes a cover configured to couple to the support member. The cover includes a substantially continuous surface configured to engage the first side of the base member at a first location that is beneath the mounting arm, cover the second side of the base member, and engage the base member at a second location that is above the mounting arm. The handrail includes a lighting element coupled to the base member within the channel such that light emitted by the lighting element is directed away from the second side of the base member. The channel is configured such that the lighting element can be installed in and removed from the base member while the handrail is installed on the mounting rail and while the cover is coupled to the support member.

[0005] Additional features and advantages will be set forth in the detailed description which follows, and, in part, will be readily apparent to those skilled in the art from the description or recognized by practicing the embodiments as described in the written description and claims thereof, as well as the appended drawings. It is to be understood that both the foregoing general description and the following detailed description are exemplary.

[0006] The accompanying drawings are included to provide further understanding and are incorporated in and constitute part of the specification. The drawings illustrate one or more embodiment, and together with the description serve to explain the principles and operation of various embodiments.

[0007] Alternative exemplary embodiments relate to other features and combinations of features as may be generally recited in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:

[0009] FIG. 1 depicts a perspective view of a handrail system, according to an exemplary embodiment;

[0010] FIG. 2 depicts a cross-sectional perspective view of the handrail system of FIG. 1, taken along line 2-2, according to an exemplary embodiment;

[0011] FIG. 3 depicts an exploded cross-sectional view of the handrail system of FIG. 1, taken along line 2-2, according to an exemplary embodiment;

[0012] FIG. 4 depicts a cross-sectional view of the handrail system of FIG. 1 mounted to a mounting surface, according to an exemplary embodiment;

[0013] FIG. 5 depicts an isolated perspective view of an endcap of the handrail system of FIG. 1, according to an exemplary embodiment;

[0014] FIG. 6 depicts an exploded perspective view of the handrail system of FIG. 6, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0015] Generally, a handrail that includes a lighting element for providing ambient or accent lighting in a hallway or room is described. The handrail includes an independent wall-facing channel configured to receive and retain a lighting element. The wall-facing channel allows users to install the lighting element prior to installing the handrail to a wall or after the handrail is installed without disrupting the finish of the handrail facing away from the wall. In some

embodiments, the wall-facing channel is formed in a separate component than a cover of the handrail, allowing users to replace the handrail cover without disrupting the lighting element. These and other advantages will be discussed in relation to the exemplary embodiments provided below.

[0016] Referring to the FIG. 1, a handrail **10** is shown, according to an exemplary embodiment. As shown in FIG. 1, the handrail **10** is attached to a mounting surface **11**, such as a wall of a building. The handrail **10** includes a first endcap **12**, a second endcap **14**, and a handrail cover **16** that is disposed between the first endcap **12** and the second endcap **14**. The endcaps **12**, **14** may also be referred to as “returns.” As seen in the embodiment depicted in FIG. 1, the endcaps **12**, **14** and cover **16** define a substantially continuous surface along longitudinal axis **a** having corresponding curvatures going from the top of the handrail system **10** to the bottom of the handrail **10** (with respect to the orientation shown in FIG. 1). In embodiments, the cover **16** and the endcaps **12**, **14** each comprise a plastic material. In particular, the cover **16** and endcaps **12**, **14** are preferably made from an impact resistant plastic. In an embodiment, the cover **16** is made from PVC. Further, in embodiments, the cover **16** is an extruded component. In embodiments, the endcaps **12**, **14** are made from PVC or ABS. In embodiments, the endcaps **12**, **14** are molded, e.g., injection molded, or made through additive manufacturing, e.g., 3D-printing.

[0017] FIG. 2 provides a perspective and partial-sectional view of the handrail **10**, showing the rear, wall-facing side of the handrail **10**. The cover **16** is snapped over a handrail support member **18**, which will be discussed in more detail in relation to FIG. 3. Additionally, the endcap **12** engages an end of the handrail support member **18**, as will be discussed more fully below in relation to FIG. 5.

[0018] Referring to FIG. 2, the support member **18** includes a central strip **20** that, when installed, is substantially parallel to the mounting surface **11**. One or more mounting arms **22** extend substantially perpendicularly from a first lateral side **19** of the central strip **20** that faces toward the mounting surface **11**, when installed. In some embodiments, a cross-section in the plane perpendicular to the direction in which mounting arm **22** extends of mounting arm **22** has a substantially circular perimeter. The diameter of the substantially circular perimeter of mounting arm **22** increases as mounting arm **22** extends away from central strip **20**. Mounting arm **22** includes a mounting end **24**. Mounting end **24** is configured to abut mounting surface **11** when installed.

[0019] As shown in FIG. 2, a lighting assembly **25** is coupled to a support member **18**. Lighting assembly **25** is positioned on a side of support member **18** that faces mounting surface **11** when installed such that lighting assembly **25** emits light toward mounting surface **11**.

[0020] Turning to FIG. 3, an exploded, cross-sectional view of the support member **18** and the cover **16** taken along the longitudinal axis **a** is shown. In embodiments, the support member **18** is made from extruded aluminum. Advantageously, each of the structures shown in the cross-sectional view of FIG. 3, and that will be discussed below, can be formed directly in the extrusion process. In other embodiments, the support member **18** can be made from another material, such as a different metal, plastic, or wood.

[0021] As shown in FIG. 3, lighting assembly **25** includes a lighting element **26** and a lighting cover **27**. In a specific embodiment, lighting element **26** is an LED strip including a plurality of light emitting diodes (LEDs). In some embodiments, lighting element **26** includes a plurality of light emitting features such that lighting element **26** emits light from a plurality of discrete locations. In other embodiments, lighting element **26** is a singular linear lighting element such that light is emitted from a continuous length of lighting assembly **25**. Lighting cover **27** is a lens formed from a clear PVC material. In other embodiments, lighting cover **27** can be formed from clear polycarbonate, other clear plastic materials, or any other transparent or translucent material.

[0022] Support member **18** includes a lighting channel **29** configured to receive and retain lighting element **26** and lighting cover **27**. Lighting channel **29** includes protrusions **31** that extend into

lighting channel **29** and retain lighting element **26** and lighting cover **27** in lighting channel **29**. In some embodiments, lighting cover **27** is flexible such that lighting cover **27** snaps into lighting channel **29** between protrusions **31** and retaining edges **33**. Protrusions **31** prevent lighting cover **27** from being pushed further into lighting channel **29** and/or from contacting lighting element **26**. Lighting channel **29** is positioned on a side of support member **18** that faces toward mounting surface **11** when installed. Referring back to FIG. **2**, lighting assembly **25** and lighting channel **29** extend substantially the entire length of support member **18** in a direction parallel to longitudinal axis **a**. Lighting channel **29** is angled such that lighting channel **29** opens toward mounting surface **11** and/or in a partially downward direction. This configuration results in the light emitted by lighting assembly **25** being directed toward mounting surface **11** and/or toward the ground below handrail **10**. In a specific embodiment, a majority of the light emitted by lighting assembly **25** is incident on mounting surface **11**, rather than the ground below handrail **10**, such that light assembly **25** provides ambient and/or accent lighting when installed in a hallway or room and is not used for emergency wayfinding lighting or direct-to-floor wayfinding lighting. Lighting cover **27** can be translucent such that lighting assembly **25** produces diffuse ambient lighting or room accent lighting.

[0023] Referring to FIG. **4**, central strip **20** includes a second lateral side **21** that faces away from mounting surface **11**, when installed. A first support arm **30** and a second support arm **32** extend away from second lateral side **21** of the central strip **20**. The first support arm **30** is in line with an upper edge of mounting arm **22** such that the central strip **20**, mounting arm **22**, and first support arm **30** form a junction in the support member **18**. The second support arm **32** is in line with a lower edge of mounting arm **22** such that the central strip **20**, mounting arm **22**, and second support arm **32** also form a junction in the support member **18**. The support arms **30**, **32** are designed to provide support for the cover **16** so that the cover **16** maintains its contour when installed on the support member **18**. In furtherance of this function, a first support panel **34** is disposed at an end of the first support arm **30**, and a second support panel **36** is disposed at an end of the second support arm **32**. As can be seen in FIG. **4**, the support panels **34**, **36** are angled relative to the support arms **30**, **32** to match the contour of the cover **16** in the region in which the cover **16** contacts the support panels **34**, **36**.

[0024] The support member **18** includes a curved support section **42** on the upper side of support member **18** creating a curved surface over which the cover **16** is supported. The cover **16** includes a first curved section **46** having a curvature that substantially matches the curvature of the curved support section **42** and that hooks over the curved support section **42**, and a first edge **48** of the cover **16**.

[0025] On a lower of the support member **18** opposite to the curved extension **38** is a lower extension **52**. Lower extension **52** forms lighting channel **29**. Further, extending from the lower extension **52** below lighting channel **29** is a ledge **56**. As shown in FIGS. **3** and **4**, the cover **16** includes a second curved section **60** with a second edge **62**. The second curved section **60** is configured to slide over the front face of and around the ledge **56** such that the second edge **62** abuts or is in close proximity to ledge **56** near the opening of lighting channel **29**.

[0026] Between the first curved section **46** and the second curved section **60** of the cover **16** is a contoured section **64**. The contoured section **64** can be of any of a variety of suitable contours available for handrail designs. In a specific embodiment, first curved section **46**, second curved section **60**, and contoured section **64** of cover **16** are configured such that handrail **10** has a curved upper section that extends closer to mounting surface **11** than the central strip **20**. In some embodiments, the curvature of cover **16** is convex near first curved section **46** and second curved section **60** and concave for at least a portion of contoured section **64**. As mentioned above, the support panels **34**, **36** maintain the contour of the contoured section **64**. Further, in embodiments, the cover **16** is formed in such a way and from such a material that the cover **16** is flexed to fit it over the curved support section **42** and the ledge **56**. In this way, the cover **16** will be in tight

engagement with the curved support section **42**, the support panels **34**, **36**, and the ledge **56**. Advantageously, the tight engagement diminishes any wiggle room or play between the cover **16** and the support member **18**.

[0027] Cover **16** is configured to be a separate, discontinuous component from support member **18** such that cover **16** can be attached to and/or removed from support member **18** while lighting assembly **25** is installed within support member **18**. As such, lighting channel **29** is separate from a handrail portion of support member **18** over which cover **16** extends, when cover **16** is attached to support member **18**. Lighting channel **29** is positioned above a bottom end of cover **16** such that lighting assembly **25** can be installed in or removed from support member **18** while cover **16** is attached to support member **18**. As such, lighting assembly **25** can be removed, replaced, and/or repaired without disrupting the visible finish of handrail system **10**. Lighting assembly **25** can also be installed in support member **18** before installing handrail system **10** to mounting surface **11** or before attaching cover **16** to support member **18**.

[0028] With reference now to FIG. 5, endcap **12** is shown. Endcap **14** is substantially similar to endcap **12** but is a mirror inverse structure. Thus, discussion of endcap **12** applies as well to endcap **14**. The endcaps **12**, **14** are configured to mate with the support member **18**. As shown in FIG. 4, the endcap **12** includes an engagement section **66** having a curved support section **72** configured to be retained within curved support section **42** of the support member **18**.

[0029] Further, the endcap **12** also includes a contoured ridge **74** that extends substantially perpendicularly from a side surface **76** of the endcap **12**. The contoured ridge **74** is similar in function to the support panels **34**, **36** and to the ledge **56** of the support member **18** as shown in FIG. 3. That is, the contoured ridge **74** supports the contoured section **64** of the cover **16** and provides a structure for the second edge **62** of the cover **16** to slide over and around. Thus, the contoured ridge **74** extends along the side of endcap **12** from a location proximal to the curved support section **72** downwardly to the bottom (with respect to the orientation shown in FIG. 4) of endcap **12**, curves toward the mounting surface **11** and then back upwardly to form a central ridge **78**, leading to the curved support section **72**.

[0030] The endcap **12** includes multiple structures configured to engage the support member **18**. Extending from the side surface **76** of the endcap **12** are a first plank **82** and a second plank **84**. The first plank **82** extends from a first side of the central ridge **78**, and the second plank **84** extends from a second side of the central ridge **78**. In this way, the planks **82**, **84** are separated by at least a distance corresponding to the thickness of the central ridge **78**. In this way, the planks **82**, **84** define a slot **86** adapted to receive the central strip **20** of the support member **18**. In a specific embodiment, first plank **82** includes a first upper arm **88** and a first lower arm **90** that are spaced vertically from each other such that there is a gap between first upper arm **88** and first lower arm **90**. Similarly, second plank **84** includes a second upper arm **92** and a second lower arm **94** that are spaced vertical from each other such that there is a gap between a second upper arm **92** and a second lower arm **94**.

[0031] As can be seen in FIG. 5, the endcap **12** has an arcuate structure. Specifically, the endcap **12** extends in an arc through an angle of about 90°. In this way, the endcap **12** caps the cover **16** and support member **18**, which are arranged parallel to the mounting surface, and transitions the end of the handrail system **10** into the wall. However, in other embodiments the endcaps **12**, **14** can instead be inside or outside corners that join sections of the handrail system **10**. That is, where two mounting surfaces meet, a first support member **18** and cover **16** can be installed on one mounting surface **11**, and a second support member **18** and cover **16** can be installed on the other mounting surface **11**. The two sections can be joined at the corner where the mounting surfaces meet, e.g., an inside corner where the angle between the mounting surfaces **11** is less than 180° or an outside corner where the angle between the mounting surfaces **11** is greater than 180°. The endcap **12** or endcap **14** will be angled to match the angle of the inside or outside corner and will have two engagement sections **66** at opposite ends of the endcap **12** or **14** to facilitate connection to both

sections of the handrail system **10**.

[0032] Referring to FIG. **6**, an exploded, cross-sectional view of handrail system **10** is shown according to an exemplary embodiment. Lighting assembly **25** includes a power cord **28** that extends from lighting element **26**. Power cord **28** is configured to direct electrical energy to lighting element **26** from a power source, such as the electrical system of a building. As shown in FIGS. **5** and **6**, endcap **12** includes a conduit **96** that allows power cord **28** to pass through endcap **12** and into mounting surface **11**. This configuration is advantageous because it allows for power cord **28** to extend from an end of lighting element **26** and into mounting surface **11** while being concealed within handrail system **10** and not be exposed to passersby.

[0033] The handrail system **10** as described herein is simpler to install than other conventional designs. In order to install the handrail system **10** on a mounting surface **11**, the endcaps **12**, **14** may first be attached to the support member **18**. As shown in FIG. **6**, the planks **82**, **84** are placed on either side of the central strip **20**. After securing the endcaps **12**, **14** to the support member **18**, the support member **18** is connected to the mounting surface **11** as shown in FIG. **4**. A fastener **104** is inserted through a hole in the central strip **20** of the support member **18**. The hole may be drilled at the time of installation, or the hole may be predrilled. The fastener passes through a hole in mounting arm **22**. The fastener **104** is then secured to a threaded structure **106** in the mounting surface **11**. The threaded structure **106** can be a threaded insert embedded into a wall, a drywall anchor, a toggle bolt, a molly, etc. The number of fasteners **104** used to secure the support member **18** to the mounting surface **11** will depend, at least in part, on the length of the support member **18**. For example, a fastener **104** may be placed every one to two feet along the length of the support member **18** to secure the support member **18** to the mounting surface **11**. While the installation method described involves attaching the endcaps **12**, **14** to the support member **18** prior to attaching the support member **18** to the mounting surface **11**, in other embodiments the endcaps **12**, **14** can be attached after the support member **18** after attaching the support member **18** to the mounting surface **11**.

[0034] Once the support member **18** and endcaps **12**, **14** are secured to the mounting surface **11**, the cover **16** is snapped onto support member **18** and endcaps **12**, **14**. In an embodiment, the second edge **62** of the second curved section **60** of the cover **16** is hooked under the ledge **56**. The first edge **48** of the first curved section **46** is then flexed over the curved support section **42**. As the first edge **48** extends over curved support section **42**, the cover **16** recovers from its flexed state to a relaxed state, which provides tight conformance of the cover **16** to the support member **18** and the endcaps **12**, **14** and completes the installation.

[0035] Thus, the installation of the handrail system **10** can be completed in three steps: (1) attaching the endcaps **12**, **14** to the support member **18**; (2) attaching the support member **18** to the mounting surface **11**; and (3) attaching the cover **16** to the support member **18**. Further, a basic installation only requires four components: the two endcaps **12**, **14**, the support member **18**, and the cover **16**. Conventional handrail designs often require additional steps and a greater number of components for the installation. For example, many handrail designs require a mounting bracket be installed on the wall. In such designs, a retainer strip for the endcaps and cover is provided and attached to the mounting bracket. Thus, these designs incorporate at least one additional component and an additional installation step.

[0036] It should be understood that the figures illustrate the exemplary embodiments in detail, and it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

[0037] Further modifications and alternative embodiments of various aspects of the invention will be apparent to those skilled in the art in view of this description. Accordingly, this description is to be construed as illustrative only. The construction and arrangements, shown in the various exemplary embodiments, are illustrative only. Although only a few embodiments have been

described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter described herein. Some elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. The order or sequence of any process, logical algorithm, or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes and omissions may also be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present invention.

[0038] For purposes of this disclosure, the term “about,” when referring to a length or distance (e.g., a length of about 10 inches), means within 10 percent above or below the referenced value (e.g., between 9 and 11 inches).

[0039] For purposes of this disclosure, the term “coupled” means the joining of two components directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional member being attached to one another. Such joining may be permanent in nature or alternatively may be removable or releasable in nature.

[0040] While the current application recites particular combinations of features in the claims appended hereto, various embodiments of the invention relate to any combination of any of the features described herein whether or not such combination is currently claimed, and any such combination of features may be claimed in this or future applications. Any of the features, elements, or components of any of the exemplary embodiments discussed above may be used alone or in combination with any of the features, elements, or components of any of the other embodiments discussed above.

[0041] In various exemplary embodiments, the relative dimensions, including angles, lengths and radii, as shown in the Figures are to scale. Actual measurements of the Figures will disclose relative dimensions, angles and proportions of the various exemplary embodiments. Various exemplary embodiments extend to various ranges around the absolute and relative dimensions, angles and proportions that may be determined from the Figures. Various exemplary embodiments include any combination of one or more relative dimensions or angles that may be determined from the Figures. Further, actual dimensions not expressly set out in this description can be determined by using the ratios of dimensions measured in the Figures in combination with the express dimensions set out in this description. As used herein, the article “a” is intended to include one or more component or element and is not intended to be construed as meaning only one.

Claims

1. A handrail configured to be installed on a mounting wall, the handrail comprising: an elongate support member having a first longitudinal end, a second longitudinal end, and a longitudinal axis extending through the first longitudinal end and the second longitudinal end, the support member comprising: a base member having a first side that faces toward the mounting wall when the support member is installed and a second side that faces away from the mounting wall when the support member is installed, the base member including a channel formed in the first side and extending in a direction parallel to the longitudinal axis; and a mounting arm extending from the first side of the base member, the mounting arm configured to engage the mounting wall when the support member is installed; a cover configured to couple to the support member, the cover comprising a substantially continuous surface configured to engage the first side of the base

- member at a first location that is beneath the mounting arm, cover the second side of the base member, and engage the base member at a second location that is above the mounting arm; and a lighting element coupled to the base member within the channel such that light emitted by the lighting element is directed away from the second side of the base member; wherein the base member is a separate, discontinuous component from the cover such that the cover can be coupled to and removed from the base member while the lighting element is coupled to the base member.
2. The handrail of claim 1, wherein the base member further comprises a central strip, a curved extension that extends from an upper portion of the central strip in a direction toward the mounting wall when the support member is installed, and a curved support section that extends from the curved extension in a direction away from the mounting wall when the support member is installed.
 3. The handrail of claim 2, wherein the substantially continuous surface of the cover defines an upper curved section, wherein the curved support section comprises a curved surface configured to engage and support the upper curved section, and wherein a curvature of the curved surface substantially matches a curvature of the upper curved section.
 4. The handrail of claim 3, wherein a portion of the cover below the upper curved section has a concave curvature.
 5. The handrail of claim 1, wherein the cover is comprised of impact resistant plastic that allows the cover to snap onto the support member during assembly.
 6. The handrail of claim 1, wherein the support member includes aluminum.
 7. The handrail of claim 1, wherein the support member includes a material from the group consisting of metal, wood, and plastic.
 8. The handrail of claim 1, the lighting element including a plurality of light emitting diodes.
 9. The handrail of claim 1, wherein the lighting element is a singular linear lighting element.
 10. The handrail of claim 1, wherein the handrail emits accent room lighting.
 11. A handrail configured to be installed on a mounting wall, the handrail comprising: an elongate support member having a first longitudinal end, a second longitudinal end, and a longitudinal axis extending through the first longitudinal end and the second longitudinal end, the support member comprising: a base member having a first side that faces toward the mounting wall when the support member is installed and a second side that faces away from the mounting wall when the support member is installed, the base member including a channel that extends in a direction parallel to the longitudinal axis; and a mounting arm extending from the first side of the base member, the mounting arm configured to engage the mounting wall when the support member is installed; a cover configured to couple to the support member, the cover comprising a substantially continuous surface configured to engage the first side of the base member at a first location that is beneath the mounting arm, cover the second side of the base member, and engage the base member at a second location that is above the mounting arm; and a lighting element coupled to the base member within the channel, the lighting element including a power cord for directing electrical energy to the lighting element from a power source; an endcap having a first endcap end and a second endcap end, the first endcap end being coupled to the first longitudinal end of the support member, and the endcap forming an arcuate shape that defines an arc between the first endcap end and the second endcap end; wherein the power cord of the lighting element passes through the endcap from the lighting element to the mounting wall.
 12. The handrail of claim 11, wherein the base member is a separate, discontinuous component from the cover such that the cover can be coupled to and removed from the base member while the lighting element is coupled to the base member.
 13. The handrail of claim 11, wherein the channel is formed in the first side of the base member such that light emitted by the lighting element is directed away from the second side of the base member; and wherein the handrail is configured to emit accent room lighting.
 14. The handrail of claim 11, wherein the lighting element is a singular linear lighting element.
 15. The handrail of claim 11, wherein the endcap is comprised of impact resistant plastic.

- 16.** The handrail of claim 11, wherein the channel is configured such that the lighting element can be installed in and removed from the base member while the handrail is installed on the mounting wall and while the cover is coupled to the support member.
- 17.** A handrail configured to be installed on a mounting wall, the handrail comprising: an elongate support member having a first longitudinal end, a second longitudinal end, and a longitudinal axis extending through the first longitudinal end and the second longitudinal end, the support member comprising: a base member having a first side that faces toward the mounting wall when the support member is installed and a second side that faces away from the mounting wall when the support member is installed, the base member including a channel formed in the first side and extending in a direction parallel to the longitudinal axis; and a mounting arm extending from the first side of the base member, the mounting arm configured to engage the mounting wall when the support member is installed; a cover coupled to the support member, the cover comprising a substantially continuous surface configured to engage the first side of the base member at a first location that is beneath the mounting arm, cover the second side of the base member, and engage the base member at a second location that is above the mounting arm; and a lighting element coupled to the base member within the channel such that light emitted by the lighting element is directed away from the second side of the base member; wherein the channel is configured such that the lighting element can be installed in and removed from the base member while the handrail is installed on the mounting wall and while the cover is coupled to the support member.
- 18.** The handrail of claim 17, wherein an upper portion of the base member defines a curved support section, wherein a portion of the cover engages the curved support section, and wherein a curvature of the curved support section matches the curvature of the portion of the cover that engages the curved support section.
- 19.** The handrail of claim 17, wherein the base member is a separate, discontinuous component from the cover such that the cover can be coupled to and removed from the base member while the lighting element is coupled to the base member.
- 20.** The handrail of claim 17, wherein the lighting element is a singular linear lighting element.
-